

ResolveMe Ticketing System

Developer Manual

1 Project Overview & Environment

This repository contains the source code and configuration for the **ResolveMe Ticketing System**. The project utilizes a strict Nix-based environment to guarantee reproducibility and ease of setup across machines.

1.1 Canonical Platform

The system of record for this project is the **canonical marking platform**, which uses:

- **OS:** `x86_64-linux`
- **Package Manager:** Nix (with flakes enabled)
- **Pinning:** `flake.lock`

1.2 Dependencies

All system and software dependencies are managed exclusively via Nix. You should not assume any pre-installed system software. The Python 3 environment is configured with the following key packages:

- `Django==6.0.1`
- `Faker==40.1.2`
- `coverage==7.10.7`
- `pillow==12.0.0`

2 Nix Entrypoints & Usage Commands

Our `flake.nix` file defines specific, automated entrypoints to ensure examiners and developers can run, test, and reset the project without manual configuration steps.

2.1 Core Commands (Required)

1. Initialize the Project

```
nix run .#init
```

- **What it does:** Downloads the required packages, applies all necessary database migrations (`makemigrations` and `migrate`) and seeds the database with a realistic scale of mock data.
- **When to use:** On a fresh clone of the repository, or when setting up the database for the first time.

2. Run the Application

```
nix run .#run
```

- **What it does:** Starts the Django development server on port 8000.
- **Access:** The application will be accessible via a web browser at <http://localhost:8000>. Note: This is also the default command if a user simply types `nix run`.

3. Run Automated Tests

```
nix run .#tests
```

- **What it does:** Executes all automated tests non-interactively using the Python coverage package.
- **Output:** It generates a coverage report and writes an XML version of this report to `./coverage.xml`.

2.2 Utility Commands (Optional but Recommended)

4. Enter the Development Shell

```
nix develop
```

- **What it does:** Upon entering, it prints a welcome banner listing all available Nix entrypoints to guide you.

5. Seed Data

```
nix run .#seed
```

- **What it does:** Injects random, realistic data into the database so the system can be tested at scale. It is designed not to fail if the database has already been seeded.

6. Unseed Data

```
nix run .#unseed
```

- **What it does:** Wipes all user-generated and seed data from the database, giving you a clean slate without having to delete the actual database file.